Lightweight Hallucination Detection for RAG: Grounding Features, a Compact BiGRU Ensemble, and a Base-Rate-Aware Evaluation

Anurag Sharma

Department of Computer Science and Engineering (Artificial Intelligence),
Maharana Pratap Engineering College, Kanpur, India

Abstract

Retrieval-augmented generation (RAG) reduces unsupported statements in model answers but does not eliminate them, and the detectors that catch the remainder usually rely on large fine-tuned encoders or paid model calls. We ask how much of that signal a small recurrent network can recover when trained from scratch on a laptop CPU. The detector is a bidirectional GRU that reads each answer sentence together with retrieved evidence. Its inputs are inexpensive grounding features and eight features from a pretrained entailment verifier, and it is trained with a token-level span loss and a response-level noisy-OR loss. On RAGTruth, with source-disjoint training, validation and test splits, a ten-seed ensemble reaches a response-level F1 of 0.739 (95% bootstrap interval 0.715–0.764), an AUROC of 0.873 and an AUPRC of 0.804; single models average an F1 of 0.726 ± 0.007. The best F1 among ensemble sizes (0.745, eight seeds) was identified on the test set and is not used as the estimate. Grounding features and seed ensembling gave the clearest gains, the verifier features a small one, and cross-attention, a larger sentence encoder and synthetic hard negatives gave none. Detection is strongest on data-to-text and weakest on summarisation, and F1 varies strongly with the hallucination rate of the generating model. A text-blind generator–task prior reaches a pooled AUROC of 0.850, so pooled AUROC alone does not establish grounding; within generator–task cells the ensemble reaches 0.749 (95% CI 0.722–0.776). It remains below published fine-tuned encoders and serves as a documented reference point.

Keywords: hallucination detection; retrieval-augmented generation; RAGTruth; grounding features; recurrent neural networks; lightweight models; faithfulness

1 Introduction

Language models are routinely asked to answer questions, summarise documents and write reports from material they are given. In retrieval-augmented generation (RAG) [1], that material is a set of retrieved passages and the model is told to stay within it. It often does not, and the deviations are usually small: a number is altered, a named entity appears from nowhere, or a caveat hardens into a certainty. Such statements are called hallucinations when the source does not support them or contradicts them [2, 3]. In customer support, health or legal settings a single wrong detail can mislead the reader, so an automatic check on every answer is attractive.

The detectors that perform best on public benchmarks are built on large pretrained models, either fine-tuned encoders [4, 5] or prompted language models [6]. Both need a GPU or a paid API, which puts them out of reach for many small teams, students and on-device applications. We therefore work in a computationally constrained setting and ask how much detection signal a small recurrent network can capture when it is trained from scratch and given explicit, inexpensive evidence of where an answer departs from its source. The system is not free of pretrained components: a sentence encoder and an entailment verifier supply input features. The recurrent network and its word embeddings, however, are not initialised from pretrained weights.

None of the ingredients is new; recurrent encoders, cross-attention, lexical-overlap features, noisy-OR aggregation and seed ensembling are all standard. The contribution lies in a specific combination, a careful evaluation, and an account of which design choices mattered:

1. **Grounding features.** A set of cheap, label-free features that compare an answer sentence with its source: exact and stemmed matches, normalised numbers and dates, n-gram overlap, entities absent from the source, and sentence-embedding similarity (Section 2.3).
2. **A compact detector and evaluation protocol.** A bidirectional GRU with a sentence-context layer, trained with token-level and response-level losses and used as a ten-seed ensemble, evaluated on RAGTruth with source-disjoint splits, a leakage check, multiple seeds and bootstrap intervals (Sections 2.4 and 2.6).
3. **Component comparisons and a base-rate analysis.** Grounding features and seed ensembling gave the clearest gains and the verifier features a small one, whereas cross-attention, a larger sentence encoder and

synthetic hard negatives did not help; the effect of the response-level objective could not be isolated. We also show how generator and task base rates shape pooled scores (Section 3).

A single model reaches a response-level F1 of 0.726 (mean over ten seeds) and the ten-seed ensemble 0.739. This is below the 0.77–0.79 reported for fine-tuned encoder detectors and above the figures reported for prompted GPT-4 and for Luna, although the published numbers are not directly comparable with ours (Section 3.6).

1.1 Related Work

Hallucination and faithfulness. Surveys define hallucination as generated content that is unfaithful to a provided source or not grounded in the world [2], and work on summarisation separates faithfulness to the input from factual correctness in general [3]. We are concerned with the former: whether an answer is supported by the passages given to the generator. Reference-free methods such as SelfCheckGPT [7] detect inconsistency by comparing several sampled answers. That multiplies the generation cost and ignores the retrieved passages, which are the natural reference in a RAG setting.

Benchmarks and detectors for RAG. RAGTruth [6] collects responses from several language models on question answering, data-to-text writing and summarisation, with human span-level hallucination labels. Detectors evaluated on it include prompted large models, Luna, a fine-tuned DeBERTa-large encoder [4], and LettuceDetect, a ModernBERT-based token classifier [5]. All of them rest on large pretrained encoders or language models, which motivates the question posed here. Lighter alternatives exist. Provenance [8] scores the factual consistency of RAG output with compact open-source NLI models; RAGulator [9] trains lightweight out-of-context detectors and reports a fine-tuned DeBERTa model as the best of those compared; and HalluGuard [10] is a 4B-parameter small reasoning model that classifies document–claim pairs as grounded or hallucinated. These systems still build on pretrained transformer or NLI models, whereas here the recurrent network is trained from scratch and pretrained models only compute input features. Entailment and alignment models such as AlignScore [11] judge whether a claim follows from a text and are a natural source of features for a detector like ours.

Sentence embeddings. Pretrained sentence encoders [12–14] give a cheap estimate of how close two sentences are in meaning. We use one as a single feature group and test whether a much larger encoder improves on it.

2 Methods and Materials

This section describes the final model and the evaluation protocol. Earlier configurations explored during development are summarised in Appendix B; they are not components of the final model.

2.1 Data

We use RAGTruth [6]. Carving a validation set out of the official training split leaves 2,264 training, 251 validation and 450 test source documents (the first two add up to the 2,515 sources of the official training split). Validation sources are whole groups of sources with identical content, stratified by task. An automatic check confirms that no source id and no duplicated source content occurs in more than one split. The test split contains 2,700 responses (900 per task and 450 from each of six generating models). With six responses per source, the training and validation sets contain 13,584 and 1,506 responses. The response-level hallucination rate is 0.445 in the training data (derived from the logged positive weight of 1.25), 0.452 on validation and 0.349 on test (943 of 2,700 responses).

2.2 Problem Formulation and Evidence Retrieval

Let D be a source (retrieved passages, a structured record or a news article) and R a response written from it, split into sentences $s_1, \ldots, s_n$. Annotators have marked character spans of R as hallucinated; a sentence is positive if it overlaps a marked span, and a response is hallucinated if at least one of its sentences is. Sentence probabilities $p_i = P(s_i \text{ is hallucinated} \mid s_i, D)$ are therefore combined with a noisy-OR,

$$f(R,D) = 1 - \prod_{i=1}^{n} (1 - p_i), \tag{1}$$

and the response is flagged when $f(R,D) \geq \tau$. The noisy-OR is a smooth substitute for $\max_i p_i$. It treats hallucination events in different sentences as independent given their inputs, which is only an approximation

Table 1. Input features of the final model.

Group	What it checks	Level
Basic	Word found in the source; word found in the evidence; number or capitalised word absent from the source; IDF weight	Token
`stem`	Stemmed word found in the whole source and in the evidence	Token, sentence
`num`	Number found in the source after normalisation (`1,234` = `1234`, `1.2 million` = `1200000`, `five` = `5`, `45%` = `0.45`), with a rounding tolerance	Token, sentence
`date`	Month or weekday name absent from the source; ISO and slash dates are parsed	Token, sentence
`ngram`	Token lies inside a bigram or trigram of stems that also occurs in the source	Token, sentence
`entity`	Run of capitalised words (for example "New York") that never occurs in the source	Token, sentence
`sbert`	Cosine similarity between the answer sentence and source sentences (maximum and mean of the top three), from the pretrained all-MiniLM-L6-v2 encoder [12,13]	Sentence
Verifier	Eight features from a frozen pretrained entailment (NLI) cross-encoder, `cross-encoder/nli-deberta-v3-xsmall`. For each answer sentence the three best passages of about 90 words are retrieved by IDF overlap (premise = passage, hypothesis = sentence). Features: maximum and mean entailment probability, maximum and mean contradiction probability, minimum neutral probability, entailment and contradiction of the best passage, and the IDF-weighted share of the sentence covered by the best passage	Sentence

because the sentences of one response are not independent; we did not test this assumption separately. The threshold τ is chosen on validation data only.

Each source is cut into chunks of 40 tokens. A chunk is scored against an answer sentence by the IDF-weighted overlap of their words, normalised by the square root of the chunk length, and the three best chunks (at most 100 tokens) form the evidence for that sentence. IDF statistics come from training sources only.

2.3 Grounding Features

For every answer token and sentence we compute simple comparisons with the source, none of which uses a label. Table 1 lists the token-level basic features, the six grounding groups and the verifier features of the final model. Token-level features are appended to the word embeddings. The eleven sentence-level grounding values are concatenated with the pooled representation and also serve as the input of the tree baseline; the eight verifier features are appended to them, which gives nineteen sentence-level values. Four further groups (alignment coverage, negation mismatch, unseen absolute or opinion words, and unit and partial-name checks) were tried during development and are not part of the final model; no controlled ablation isolates their effect (Appendix B). The stemmer is a short suffix-stripping routine, so the grounding features can be reproduced without external language resources, whereas the `sbert` and verifier features require pretrained models.

2.4 Model Architecture

Figure 1 shows the architecture. Word embeddings (dimension 64, randomly initialised, with spatial dropout) are concatenated with the token features and passed through a bidirectional GRU [15] (64 units), which gives the answer states A. A second BiGRU encodes the evidence into states E, and cross-attention gives each answer position a context vector $C = \mathrm{softmax}(AE^\top)E$. The combination $[A; C; A \odot C; A - C]$ is projected, layer-normalised, passed through a third BiGRU and pooled by maximum and mean over positions. The pooled vector is concatenated with the standardised sentence-level features and passed through a hidden layer, producing a sentence vector z_i.

A *sentence-context layer* then lets each sentence see the rest of its response: the vectors $z_1, \ldots, z_n$ of one response go through a small BiGRU (32 units per direction), and its output is concatenated with z_i before a single logit is produced. The motivation is that a clause is often supported or contradicted by something said elsewhere in the same answer. During training, an auxiliary token head reads the states of the last word-level BiGRU and predicts, for every answer word, whether it lies inside a hallucinated span; only the sentence logits are used to score a response at test time. The word embeddings and all recurrent layers are trained from scratch, whereas the sentence-embedding and verifier features come from pretrained models and enter the network as inputs. The cost of computing them is not included in the inference timings reported below.

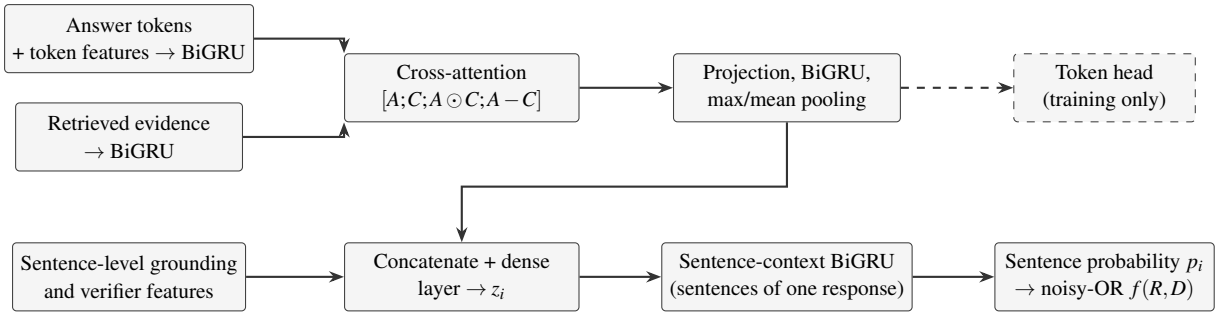


Figure 1. Architecture of one member of the final model. A and E are the answer and evidence states and C the cross-attended context. A response is flagged when the noisy-OR of its (seed-averaged) sentence probabilities exceeds a threshold chosen on validation data.

2.5 Training and Ensemble Scoring

Objectives. All three losses are weighted binary cross-entropies,

$$\mathscr{L}(w; y, q) = -\frac{1}{N}\sum_{j=1}^{N}\Big[w\, y_j \log q_j + (1-y_j)\log(1-q_j)\Big], \tag{2}$$

where $y_j \in \{0,1\}$ is the label and q_j the predicted probability (label smoothing is omitted from the expression for brevity). The sentence loss $\mathscr{L}_{\text{sent}}$ is computed over sentences with $q_j = p_i$ and positive weight $w_{\text{sent}} = \min\big((1-\pi)/\pi, 3\big)$, where π is the positive rate of training sentences. The auxiliary token loss $\mathscr{L}_{\text{tok}}$ is computed over answer words, with $y_j = 1$ for words inside a hallucinated span; only about 4% of words fall inside such spans, so its positive weight is capped at 10, and its loss weight is 0.5. The response-level loss $\mathscr{L}_{\text{resp}}$ is computed over responses with $q_j = f(R,D)$ and applied to the noisy-OR logit, also with a loss weight of 0.5. With sentence logits ℓ_i and $S = \sum_i \text{softplus}(\ell_i)$, that logit is $S + \log(1 - e^{-S})$, which equals $\text{logit}\big(1 - \prod_i (1-p_i)\big)$ and is numerically stable. Its positive weight is $\min\big((1-\rho)/\rho, 3\big)$, where ρ is the response-level positive rate of the training data. Batches are built from whole responses (about 128 sentences each).

Optimisation and selection. We use AdamW [16] with dropout, spatial and word dropout, label smoothing [17] and an exponential moving average of the weights; the averaged weights are the ones evaluated (all values in Appendix A). The learning rate is halved after two evaluations without improvement. Evaluation runs every 200 steps and at the end of each epoch. We keep the checkpoint with the best response-level validation AUPRC, stop training after eight evaluations without improvement, and set the decision threshold to maximise validation F1.

Ensemble scoring. The final model averages the sentence logits of $k = 10$ independently trained networks (seeds 0–9) that share the architecture and settings above, $\bar{\ell}_i = \frac{1}{k}\sum_{m=1}^{k} \ell_i^{(m)}$. The average is converted to $\bar{p}_i = \sigma(\bar{\ell}_i)$, and the sentences of a response are aggregated with the noisy-OR of Eq. (1). The threshold is re-selected on validation data for the ensemble (0.474 for ten seeds). A stacking stage fitted on validation responses was explored for five-seed ensembles only and is not part of the final model (Appendix B).

2.6 Evaluation Protocol

Metrics. The main metric is the F1 score of the hallucinated class at response level. We also report precision, recall, AUROC and AUPRC, overall and per task, together with word-level AUPRC and AUROC against the human span labels. Prevalence differs strongly between tasks and generators, so comparisons across them rely on AUROC and AUPRC. Confidence intervals are 95% bootstrap intervals that resample source documents; they reflect test-set sampling only. "Mean ± s.d." always denotes the sample standard deviation of single models over independently trained seeds. An ensemble is one model and carries no standard deviation; its uncertainty is given by the bootstrap interval alone.

Baselines. The baselines are (i) flagging every response; (ii) a task-type rule that flags data-to-text responses only; (iii) lexical overlap, defined as one minus the IDF-weighted share of answer words found in the evidence; and (iv) a histogram-based gradient-boosted tree model on the sentence-level features, with and without a task indicator and with the verifier features. We also include a text-blind prior that uses only the generating model and the task: each response is scored by the hallucination rate of its generator–task cell, estimated on the training responses

(the official training split without the validation sources). All thresholds, including that of the prior, are chosen on validation data.

Final model and development configurations. The final model is configuration F: the architecture above with the six grounding groups, the token and response-level losses, the sentence-context layer, no synthetic data, the standard evidence length (top three chunks) and eight verifier features. It was trained with ten seeds (0–9; seeds 5–9 in a separate run with identical settings). Configurations A to E, trained with three or five seeds, are earlier variants (Appendix B); several differ from F in more than one setting, so they are not single-factor ablations unless stated. We preferred F to its verifier-free counterpart E on validation data, where its AUROC was higher in all five matched seeds (0.857 against 0.849 on average) and its AUPRC slightly higher (0.849 against 0.845). The test results of all variants were visible during development.

Ensemble size. We evaluated every ensemble size from one to ten (the first k seeds), re-selecting the threshold on validation data each time. The ten-seed ensemble was not singled out in advance; it is simply the all-seeds result and serves as the primary estimate. The eight-seed ensemble has the highest test F1 among the sizes, but because that choice used the test set, its F1 is optimistically selected (Section 3.2).

2.7 Implementation and Availability

All experiments used Python 3.13.7, TensorFlow 2.21.0 and NumPy 2.2.6 on a laptop CPU without a GPU (Intel Core Ultra 5 125H, 14 cores and 18 logical processors, 16 GB RAM, Windows 11). Training runs of the final configuration took 8,100 to 11,300 seconds for seeds 0–4 (logs for seeds 5–9 were not available). Runs of the same size differed in duration by a factor of two or more in our logs, so all timings are indicative. Inference for one network took 0.14–0.15 s per response, excluding sentence encoding and the verifier; by extrapolation (not separately timed), the ten-network ensemble needs roughly 1.4–1.5 s.

Data and code availability. RAGTruth is publicly available [6]. The source code, saved predictions and experimental run logs are not publicly available at the time of publication.

3 Results

3.1 Overall Performance

Table 2 gives the overall results on the test split; the ten-seed ensemble is the final model and the primary result. Over the ten seeds, single networks have a mean AUROC of 0.866 ± 0.003 and a mean AUPRC of 0.793 ± 0.005. Seeds 0–4 and seeds 5–9, trained in separate runs with identical settings, gave essentially the same mean F1 (0.725 and 0.726), and their five-seed ensembles reached 0.741 and 0.739, which suggests that the training procedure is reproducible. The ten-seed ensemble has an AUROC of 0.873 (95% CI 0.858–0.887) and an AUPRC of 0.804 (0.775–0.833).

Table 2. Response-level results on the RAGTruth test split (hallucinated class positive). "F1 over seeds" is the mean ± sample standard deviation of ten single models, each with its own validation-selected threshold. The remaining columns refer to seed 0 (single-model row) or to the ten-seed ensemble. †AUROC over pairs of responses from the same generator–task cell only (18 cells of 150 responses; cell AUROCs weighted by the number of hallucinated–faithful pairs), for which the prior has an AUROC of 0.5 by construction. Intervals for the prior and for the within-cell AUROC are 95% bootstrap intervals that resample source documents.

Model	F1 over seeds	F1 (95% CI)	P	R	AUROC	AUPRC
All hallucinated	–	0.518 (0.491–0.544)	0.349	1.000	0.500	0.349
Task-type rule	–	0.628 (0.597–0.661)	0.643	0.614	0.716	0.530
Lexical overlap	–	0.620 (0.593–0.648)	0.483	0.866	0.720	0.504
Tree model, features	–	0.675 (0.649–0.700)	0.566	0.835	0.814	0.680
Tree model, features + task	–	0.658 (0.632–0.685)	0.554	0.811	0.796	0.650
Tree model, features + verifier + task	–	0.655 (0.631–0.682)	0.528	0.861	0.803	0.662
Generator×task prior (text-blind)	–	0.666 (0.638–0.693)	0.565	0.809	0.850	0.759
Final model, single network	0.726 ± 0.007	0.722 (0.697–0.746)	0.664	0.791	0.863	0.789
Final model, ensemble of 10 seeds	–	**0.739 (0.715–0.764)**	**0.721**	**0.758**	**0.873**	**0.804**
Ensemble, within generator×task cells†	–	–	–	–	0.749 (0.722–0.776)	–

Paired bootstrap differences in F1 between the ensemble and three of the baselines are positive: 0.065 (95% CI 0.045–0.084) over the tree model on the same features, 0.085 over the tree model with verifier features and task indicator, and 0.119 over lexical overlap. Against the generator×task prior the paired differences are 0.073 in F1 (95% CI 0.050–0.099), 0.023 in AUROC (0.008–0.039) and 0.045 in AUPRC (0.017–0.071); this comparison is examined in Section 3.5. At word level, the ensemble reaches an AUPRC of 0.348 and an AUROC of 0.860 against the human span labels (chance level for AUPRC 0.042).

3.2 Training Behaviour and Ensemble Size

Validation AUROC and AUPRC rise steeply over the first 400 steps and level off between about 1,000 and 1,700 steps (mean AUPRC about 0.845–0.848, AUROC about 0.855–0.856), after which they decline slowly, to about 0.835 and 0.847 at step 2,600 (Figure 2). This indicates overfitting despite the regularisation used. The checkpoints kept by the AUPRC criterion lie between steps 834 and 1,671 (median 1,400), roughly one to two epochs. These curves are optimistic estimates of generalisation, since each seed's checkpoint was chosen on the same validation set.

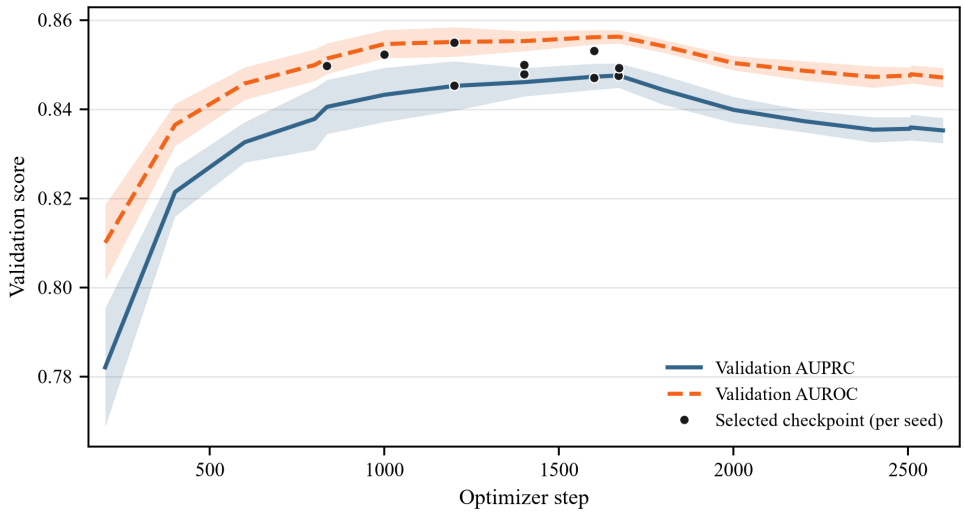


Figure 2. Response-level validation AUPRC (solid) and AUROC (dashed) of the final configuration during training, mean ± s.d. over ten seeds. Black dots mark the checkpoint kept for each seed (plotted at its validation score).

Figure 3 and Table 3 (below) report test performance for ensembles of the first k seeds. Ensembling helps clearly at first: F1 rises from 0.722 for one network to about 0.735 for two or three networks, while AUROC and AUPRC change little beyond three seeds. After that F1 is irregular. The eight-seed ensemble has the highest F1 of the ten sizes (0.745; precision 0.695, recall 0.803; threshold 0.244), and the ten-seed ensemble is lower (0.739; precision 0.721, recall 0.758; threshold 0.474). At its own threshold the eight-seed ensemble produced 757 true positives, 332 false positives, 186 false negatives and 1,425 true negatives; per-task counts for it are not available.

We do not read this difference as an advantage of eight networks. First, as rankers the two ensembles are almost identical (AUROC 0.8732 against 0.8725, AUPRC 0.8043 against 0.8038), and neither metric depends on a threshold. Second, the F1 gap of 0.006 comes from a different operating point: the validation-selected threshold moves from 0.474 to 0.244, trading precision for recall, and thresholds across sizes range from 0.21 to 0.55. Third, the half-width of the ten-seed F1 interval is about 0.025, four times the gap. Fourth, random subsets of eight seeds give an F1 of 0.738 ± 0.002 (the subsets share seeds, so this spread understates the true uncertainty), almost the ten-seed value, which makes the first-eight ensemble a favourable draw. A clean test would need new seeds or held-out data; we therefore report the ten-seed ensemble as the estimate and the eight-seed result as a post-hoc observation.

3.3 Component Comparisons

Table 4 collects the comparisons that bear on the final design. Its rows come from earlier three- and five-seed runs and differ from the final model in the ways stated, so each comparison holds within one block only. Secondary experiments (encoder size, evidence length, per-task thresholds, stacking and the development path A–F) are in Appendix B.

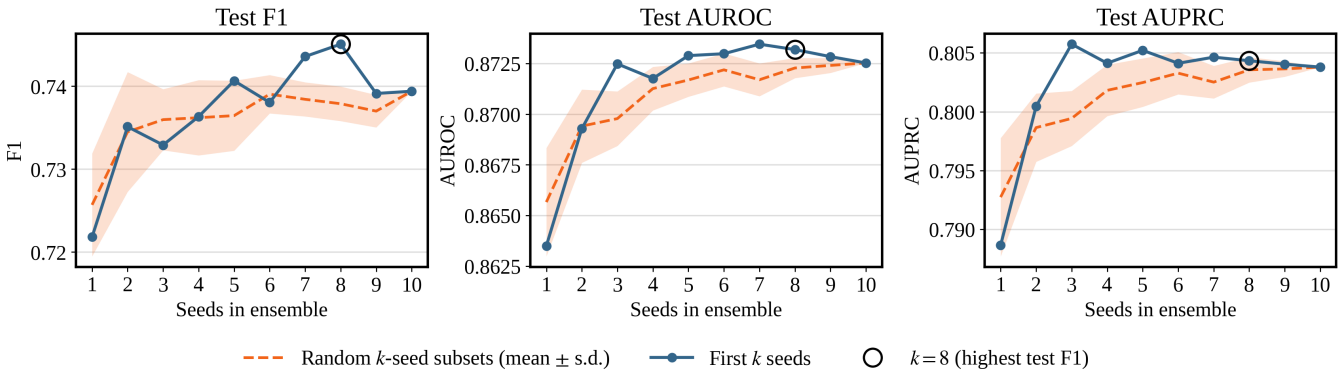


Figure 3. Test F1, AUROC and AUPRC against ensemble size. Blue: ensemble of the first k seeds (threshold re-selected on validation data for every size). Orange: mean ± s.d. over random k-subsets of the ten seeds (subsets overlap and are not independent). The ring marks $k = 8$, the size with the highest test F1; that size was identified on the test set. All ten sizes are shown.

Table 3. Test performance for selected ensemble sizes (ensemble of the first k seeds, threshold re-selected on validation data for every size; sizes not listed are shown in Figure 3). The last column gives the mean ± standard deviation of F1 over random k-subsets of the ten seeds (10–20 subsets, which overlap and are not independent; one subset for $k = 10$). The row for $k = 10$ is the final model; $k = 8$ (italic) has the highest test F1, was identified on the test set, and is not used as the estimate.

k	F1	P	R	AUROC	AUPRC	Threshold	F1, random subsets
1	0.722	0.664	0.791	0.863	0.789	0.21	0.726±0.006
2	0.735	0.691	0.786	0.869	0.800	0.25	0.734±0.007
3	0.733	0.727	0.739	0.872	0.806	0.55	0.736±0.004
5	0.741	0.735	0.747	0.873	0.805	0.55	0.736±0.004
8	*0.745*	*0.695*	*0.803*	*0.873*	*0.804*	*0.24*	*0.738±0.002*
10	**0.739**	**0.721**	**0.758**	**0.873**	**0.804**	**0.47**	**0.739±0.000**

Table 4. Single-model comparisons on the test split. F1 is mean ± standard deviation over seeds; validation AUPRC is the selection criterion, averaged over seeds; test AUROC is for seed 0. Compare rows within a block only. ‡Earlier runs not covered by the results file used for the main tables. Rows A and B differ in two settings (features and token loss); rows in the last block are variants of configuration D, which also uses synthetic negatives and longer data-to-text evidence and is not the final model.

Comparison	Variant	Seeds	Test F1	Val. AUPRC	Test AUROC
Grounding features‡	(A) basic token features, no span loss	3	0.670 ± 0.011	–	0.831
	(B) A + six feature groups + token loss	3	0.695 ± 0.014	–	0.847
Verifier features	(E) final model without verifier features	5	0.723 ± 0.003	0.845	0.863
	(F) final model	5	0.725 ± 0.005	0.849	0.863
Synthetic negatives	(E) final model without verifier features	5	0.723 ± 0.003	0.845	0.863
	E + synthetic hard negatives	3	0.706 ± 0.005	0.834	0.850
Cross-attention and token loss	(D) reference	3	0.702 ± 0.007	0.832	0.850
	D, cross-attention removed	3	0.708 ± 0.006	0.834	0.846
	D, token loss switched off	3	0.704 ± 0.003	0.831	0.845

Grounding features. Adding the six grounding groups, together with the span loss, to the basic network raised the three-seed mean F1 from 0.670 to 0.695 and the seed-0 AUROC from 0.831 to 0.847 (A to B). In earlier three-seed runs, switching the token loss on or off changed F1 by at most 0.005,‡ so this step is largely attributable to the features, although it was not isolated in a single controlled run. The same features lift a tree model to an AUROC of 0.814, against 0.720 for lexical overlap. The network still contributes beyond the features: network A, which sees only basic token features, roughly matches the tree model built on the full sentence-level grounding features (mean F1 0.670 against 0.675, seed-0 AUROC 0.831 against 0.814),‡ and on data-to-text the tree model reaches an AUROC of only 0.632 while every network variant we ran reaches at least 0.81.

Verifier features. Adding the verifier features (F against E) moved the mean F1 from 0.723 to 0.725 over five seeds (0.726 over ten), a change well inside the seed spread. The validation AUROC was higher in all five matched seeds, and five-seed ensembles gained 0.004 in AUROC (0.869 to 0.873) and 0.003 in AUPRC. The tree model with a task indicator also gained slightly in AUROC (0.803 against 0.796) and AUPRC (0.662 against 0.650) but not in F1 (0.655 against 0.658). The effect is small and mostly visible in AUROC, and it makes the detector depend on an external pretrained model.

Synthetic hard negatives (not in the final model). In configuration D, a random 30% of the faithful training responses received a corrupted copy in which one sentence was made unsupported, and the changed characters were labelled as a hallucinated span. Three deterministic heuristics produced the corruptions: shifting a number to a value absent from the source, swapping a capitalised word for another from the same source, and inserting "not" after an auxiliary verb. Validation and test data were never modified. Adding these responses to E, with nothing else changed, lowered the mean F1 from 0.723 (five-seed mean of E) to 0.706 (three seeds). All three matched seeds were lower, by 0.008 to 0.018, and so was the validation AUPRC in each matched seed; the same rows also lowered the tree baseline from an AUROC of 0.814 to 0.803. Two caveats apply. The response-level positive weight changes with the training prevalence (0.91 with synthetic data, 1.25 without), so the comparison is not a pure data effect, and with three seeds the result shows that this augmentation does not help rather than giving a precise estimate. About 2,260 synthetic responses were added (30% of roughly 7,540 faithful training responses, derived rather than read from a log); their split by heuristic was not recorded.

Cross-attention and token loss. Removing cross-attention from D changed the mean F1 from 0.702 ± 0.007 to 0.708 ± 0.006, and an earlier single-seed ablation of configuration A showed the same absence of effect (F1 0.676 against 0.677).‡ Cross-attention therefore shows no measurable benefit in our setting, yet it remains part of the final architecture, which was not retrained without it. Switching the token loss off changed the response-level F1 by $+0.002$, but the word-level scores differ sharply: the token AUPRC of D is 0.323 with the loss and 0.127 without it, against a chance level of 0.042. The token loss thus does not improve response-level scores, but it is what makes the word-level scores useful for localising unsupported text.

3.4 Task and Generator Analysis

Table 5 and Figure 4 compare the ten-seed ensemble with the baselines on each task. The ensemble is clearly ahead of lexical overlap everywhere, but its lead over the tree models is uneven. On data-to-text the lead is large (AUROC 0.859 against 0.629–0.632, AUPRC 0.909 against 0.748–0.751). On question answering the ensemble has the higher F1 (0.619 against at most 0.537) and AUPRC (0.658 against at most 0.633), but its AUROC (0.873) is no higher than that of the tree model with verifier features (0.876). On summarisation the tree model with verifier features is at least as good on all three metrics (F1 0.485, AUROC 0.754, AUPRC 0.514, against 0.468, 0.747 and 0.508), although these differences lie well inside the ensemble's bootstrap intervals. The overall AUROC gain over the tree models (0.873 against 0.814) is therefore driven mainly by data-to-text, and on summarisation the recurrent model shows no demonstrated advantage over a tree model on the same features.

Table 5. Per-task test results, F1 / AUROC / AUPRC. The all-hallucinated row also gives the prevalence of each task (its AUPRC). Per-task results for the other baselines are in Figure 4.

Model	QA	Data-to-text	Summarisation
All hallucinated	0.302 / 0.500 / 0.178	0.783 / 0.500 / 0.643	0.370 / 0.500 / 0.227
Tree model, features	0.537 / 0.847 / 0.587	0.784 / 0.632 / 0.751	0.444 / 0.705 / 0.442
Final model, ensemble of 10 seeds	0.619 / 0.873 / 0.658	0.852 / 0.859 / 0.909	0.468 / 0.747 / 0.508

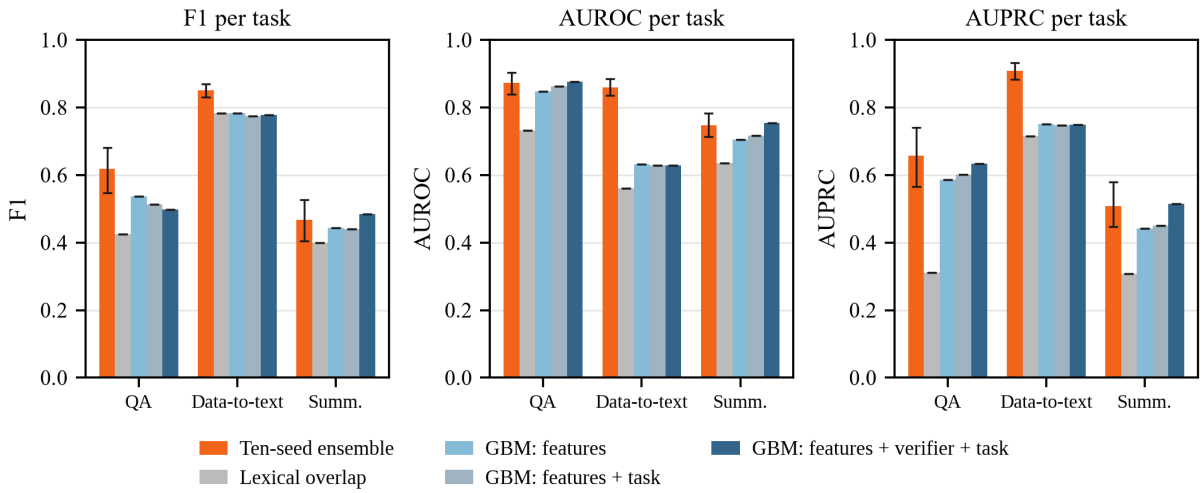


Figure 4. Per-task test F1, AUROC and AUPRC of the ten-seed ensemble and of the baselines (Summ. = summarisation). Error bars on the ensemble are 95% bootstrap intervals that resample source documents. GBM = gradient-boosted tree model, shown with features only, features plus task indicator, and features plus verifier features and task indicator. Prevalence differs by task (Table 5), so tasks should be compared by AUROC and AUPRC rather than F1.

Data-to-text responses are 64% hallucinated, so flagging everything already scores an F1 of 0.783 there, and AUROC is the better guide. Summarisation is the weakest task, and word-level localisation is also weakest there (token AUPRC 0.161, against 0.457 on question answering and 0.328 on data-to-text). Recall of the ten-seed ensemble by annotated hallucination type (types can co-occur in a response) is 0.78 for evident conflicts (459 cases), 0.81 for evident baseless information (542) and 0.84 for subtle baseless information (141). The 15 subtle conflicts are detected at 0.47 (95% Wilson interval 0.25–0.70), too few cases to interpret. At the shared threshold of 0.474 (Figure 5), most errors in data-to-text are false alarms (39% of faithful responses flagged, 10% of hallucinated ones missed), whereas 60% of hallucinated summaries are missed and only 9% of faithful summaries are flagged.

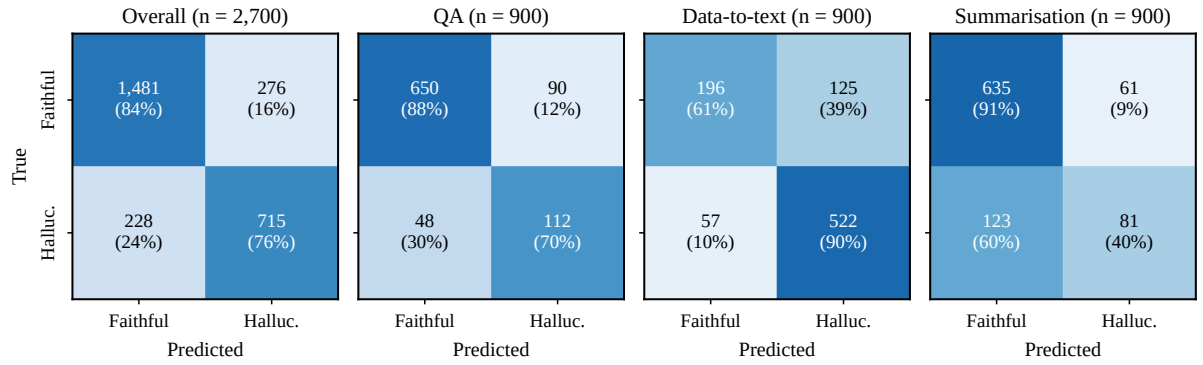


Figure 5. Confusion matrices of the final ten-seed ensemble on the RAGTruth test split (2,700 responses) at the validation-selected threshold of 0.474. Cells show counts and row percentages; rows are the true class and columns the predicted class (Halluc. = hallucinated). The per-task counts sum to the overall counts.

Overall, 715 of the 943 hallucinated responses are caught and 228 are missed (24.2%), while 1,481 of the 1,757 faithful responses are passed and 276 are wrongly flagged (15.7%); precision is 0.721 and recall 0.758. Question answering lies between the two other tasks (30% of hallucinated responses missed, 12% false alarms). The per-task F1 values derived from these counts are 0.619 (QA), 0.852 (data-to-text) and 0.468 (summarisation), as in Table 5. Because the threshold is shared across tasks and the hallucination rate differs strongly between them (0.64, 0.18 and 0.23), error rates should be read together with these base rates.

Table 6 gives the results for each generating model. F1 depends strongly on the base rate: for the two GPT models only about 10% of responses are hallucinated and F1 is 0.40–0.44 (precision 0.34–0.46), whereas for the Llama-2 and Mistral models, with 38–56% hallucinated, it is 0.71–0.81. AUROC varies less (0.79–0.89). Validation and test hallucination rates also differ (0.452 against 0.349), and the thresholds of single seeds vary from 0.16 to 0.58, so F1 rests on a poorly determined threshold. AUROC and AUPRC are therefore the more stable indicators,

and we recommend re-tuning the threshold once the deployment prevalence is known.

Table 6. Test results of the ten-seed ensemble by generating model (450 responses each, shared threshold 0.474).

Generator	Prevalence	P	R	F1	AUROC	AUPRC
gpt-3.5-turbo-0613	0.102	0.463	0.413	0.437	0.868	0.400
gpt-4-0613	0.093	0.339	0.500	0.404	0.840	0.386
llama-2-13b-chat	0.460	0.765	0.865	0.812	0.891	0.892
llama-2-70b-chat	0.380	0.702	0.842	0.766	0.863	0.791
llama-2-7b-chat	0.502	0.712	0.712	0.712	0.788	0.779
mistral-7B-instruct	0.558	0.857	0.761	0.806	0.868	0.897

3.5 Pooled Scores and the Text-Blind Prior

The hallucination rate in the test split depends strongly on the task (0.64 for data-to-text, 0.18 for question answering and 0.23 for summarisation; Table 5) and on the generator (Table 6). A rule that knows only which generator wrote a response and which task it belongs to can therefore rank responses by the base rate of their generator–task cell, and flag every response from a high-prevalence cell, without reading a word of the response. This prior is not a competing detector, since it cannot separate hallucinated from faithful responses within a cell. It matters because a text-based detector may also pick up the task and the generator's style from the response and its source, in which case pooled metrics credit it for base-rate information as well as for detecting unsupported content.

On our test split the prior reaches a pooled AUROC of 0.850 (95% CI 0.834–0.864), an AUPRC of 0.759 (0.726–0.791) and an F1 of 0.666 (0.638–0.693), against 0.873, 0.804 and 0.739 for the ten-seed ensemble (single-network means: AUROC 0.866, F1 0.726). The pooled AUROC of the prior is thus close to that of the final model; the ensemble's paired advantage is 0.023 in AUROC (95% CI 0.008–0.039), 0.045 in AUPRC (0.017–0.071) and 0.073 in F1 (0.050–0.099). Estimating the cell rates on the test labels themselves, which is not a legitimate baseline and serves only as an upper bound, gives an AUROC of 0.852 and an F1 of 0.701.

Two further points bear on how much of the ensemble's score is text-level signal. The per-task AUROCs in Table 5 remove the base-rate differences between tasks but not between generators; the per-generator AUROCs in Table 6 do the reverse, because each generator's 450 responses still mix three tasks. Within a generator–task cell the prior carries no information, so the within-cell AUROC isolates what the model reads from the text. For the final ensemble it is 0.749 (95% CI 0.722–0.776), clearly above chance but well below the pooled 0.873. The 18 cell AUROCs (Table 7) range from 0.61 to 0.93; the highest and lowest values come from cells with very few hallucinated responses (for example about 1% for gpt-4 on question answering) and are imprecise. The pooled AUROC thus blends text-level signal with base-rate information, and these numbers do not allow the two to be separated exactly.

Table 7. AUROC of the ten-seed ensemble within each generator–task cell (150 responses per cell), with the hallucination rate of the cell in parentheses. Cells with a rate of 0.04 or lower contain at most six hallucinated responses, so their AUROCs are imprecise.

Generator	QA	Data-to-text	Summarisation
gpt-3.5-turbo-0613	0.932 (0.03)	0.754 (0.25)	0.786 (0.03)
gpt-4-0613	0.611 (0.01)	0.695 (0.23)	0.669 (0.04)
llama-2-13b-chat	0.861 (0.24)	0.802 (0.92)	0.623 (0.22)
llama-2-70b-chat	0.779 (0.23)	0.685 (0.75)	0.763 (0.16)
llama-2-7b-chat	0.771 (0.35)	0.741 (0.82)	0.685 (0.34)
mistral-7B-instruct	0.879 (0.21)	0.822 (0.89)	0.710 (0.57)

3.6 Comparison with Published Results

Response-level F1 values reported for RAGTruth are 63.4 for prompted GPT-4, 65.4 for Luna, and 76.8 and 79.2 for the base and large LettuceDetect models, as collected in the LettuceDetect paper [5]. Our single-model mean (72.6) and ten-seed ensemble (73.9) lie between the first two and the last. We make no claim of parity with any of these systems. The figures come from different papers and are comparable only if the same metric (F1 of the

hallucinated class at response level), the same test split and the same treatment of implicit labels were used. We did not verify whether this holds, and a fair comparison would require re-running those systems or contacting their authors.

4 Figures and Tables

Tables and figures are placed beside the passages that first discuss them rather than collected in one place. The model architecture (Figure 1, p. 4) and the input features (Table 1, p. 3) belong to the methodology. The primary results are Table 2 (p. 5), the component comparisons in Table 4 (p. 7), and the per-task and per-generator analyses in Table 5, Figure 4 and Table 6 (pp. 8–10). Training curves (Figure 2), ensemble-size evidence (Figure 3, Table 3), confusion matrices (Figure 5) and within-cell AUROC (Table 7) accompany the sections that interpret them. Only the model settings (Table 8) and the development path (Table 9) are in the appendices.

Several conventions apply throughout. Intervals are 95% bootstrap intervals that resample source documents, and "mean $\pm$ s.d." refers to single models over seeds. A dagger (†) marks the within-cell AUROC and a double dagger (‡) marks results from earlier runs outside the main results file. The ten-seed ensemble ($k = 10$) is the primary estimate in every table and figure; the eight-seed ensemble ($k = 8$) is shown only as a post-hoc observation.

5 Discussion

This section interprets the results above. Unless stated otherwise, the interpretations are hypotheses.

5.1 Interpretation

Grounding features and the recurrent encoder. Grounding features gave the most reliable improvement for the recurrent model, and they also lifted a tree model well above lexical overlap. Much of the detection signal in this setting therefore seems to lie in explicit comparisons between answer and source, which even a small network can exploit once they are supplied. Beyond the features, the recurrent encoder adds value mainly on data-to-text; on question answering and summarisation a tree model with verifier features is comparable in AUROC. One possible reason why cross-attention had no effect, which we did not test, is that attention can only align words that share an embedding. Embeddings trained on about two thousand sources are probably too weak for that, so the exact-match flags end up carrying the alignment.

Single networks and ensembles. Averaging a few seeds raises F1, AUROC and AUPRC modestly, and the benefit saturates early: about three seeds capture most of the gain in AUROC and AUPRC, and beyond that F1 changes more with the operating point than with ensemble quality. The improvement should not be read as evidence that the ensemble grounds its decisions better in the evidence. Averaging reduces seed-specific variance in whatever the networks have learned, including base-rate and style cues, and our experiments cannot separate these. A single network (F1 0.726) or a three-seed ensemble is the more practical choice when latency matters, at a cost of roughly 0.01 to 0.02 F1.

Task differences. The error pattern in data-to-text is consistent with a task base rate of 64% combined with a threshold shared across tasks, but we did not test this explanation. In summarisation, a single incorrect clause often sits inside an otherwise well-grounded sentence, which sentence-level scores may dilute; this too is untested. That a tree model on the same features performs comparably on summarisation suggests that the added capacity of the recurrent encoder does not help on this task.

What pooled scores can and cannot show. The final ensemble beats the text-based baselines on pooled metrics, and a text-blind generator–task prior comes close in pooled AUROC (0.850 against 0.873). Within generator–task cells, where the prior is uninformative, the ensemble's AUROC is 0.749 (95% CI 0.722–0.776). The model therefore does discriminate hallucinated from faithful responses on the basis of their text, but less well than the pooled AUROC suggests, and the pooled figure should not be read as a measure of grounding ability alone. The within-cell AUROC varies across cells (0.61–0.93) and is lowest for summarisation and for GPT-4; it has not been compared with a within-cell baseline such as the tree model.

5.2 Limitations

- **Selection on the test set.** All ensemble sizes from one to ten were scored on the test set, and the eight-seed ensemble (F1 0.745, no confidence interval available) was the best; the ten-seed result is the reported estimate. More than thirty configurations were also compared while their test scores were visible. Checkpoints, thresholds and ensemble members used only training and validation data, but the choice of which configuration to report may still make the numbers optimistic. Evaluation on freshly held-out data would remove this concern.
- **Statistical power of the ablations.** Configurations A–E have three to five seeds and F has ten, and some ablations were run on D rather than on the final model. The response-level loss and the context layer were not ablated individually, and differences below roughly 0.01–0.02 F1 should be treated as noise.
- **Small validation set.** The same 251 validation sources serve for checkpoint selection and threshold choice, and their hallucination rate differs from that of the test set.
- **Pretrained components.** The final model depends on features from a pretrained entailment model (an NLI cross-encoder run once on CPU and cached) and on a pretrained sentence encoder. The reported timings exclude both, so the low cost of the recurrent network is not the cost of the full pipeline.
- **Base rates.** Pooled metrics partly reflect generator and task prevalence. We report the within-cell AUROC (0.749) and the generator–task prior, but no within-cell result is available for the baselines or for other configurations.
- **Features and scope.** Stem, n-gram and entity features cannot recognise a correct paraphrase or a subtle contradiction. The evaluation covers one benchmark, one language (English) and three task types, so performance elsewhere is unknown.
- **Earlier runs.** Configurations A and B, and the results marked with a double dagger, come from earlier runs that are not part of the results file behind the main tables.

5.3 Deployment and Future Work

A missed hallucination lets an unsupported statement reach the reader, and the final ensemble misses 60% of hallucinated summaries (Figure 5). A false alarm causes unnecessary regeneration or review, and in data-to-text the same threshold flags 39% of faithful responses. Because performance and the appropriate threshold vary with task and generator (Table 6), the detector should be recalibrated for the deployment setting and used as a screening aid ahead of human review, not as a guarantee of faithfulness, particularly in health, legal and similar high-stakes uses. The intended pipeline (split the draft answer into sentences, compute features against the retrieved passages, score each sentence, then keep, mark, regenerate or remove it according to validation-tuned thresholds) has not been evaluated. Future work includes comparing plain RAG with the checked pipeline on hand-verified questions and evaluating the token head as a highlighter of unsupported words.

6 Conclusion

This study asked how far a compact bidirectional GRU, trained from scratch on a laptop CPU, can detect hallucinations in RAG outputs when it receives inexpensive grounding features and eight verifier features. On RAGTruth with source-disjoint splits, the final ten-seed ensemble reaches a response-level F1 of 0.739, an AUROC of 0.873 and an AUPRC of 0.804, while single networks average an F1 of 0.726. The eight-seed ensemble has a higher test F1 (0.745), but that size was identified on the test set and the difference lies within the noise. Grounding features gave the clearest gain; cross-attention, a larger sentence encoder and synthetic negatives gave no benefit.

The detector remains below published fine-tuned encoders. It is weakest on summarisation, where the final ensemble misses 60% of the hallucinated responses; it raises many false alarms on data-to-text; and its pooled AUROC (0.873) is close to that of a prior based only on generator and task (0.850), with a within-cell AUROC of 0.749. The results are best read as a documented reference point for a small CPU-trained recurrent detector. The main open questions are a test of the final configuration on new data or seeds, controlled ablations of the final model, and an end-to-end test inside a RAG pipeline.

References

[1] P. Lewis *et al.*, "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Advances in Neural Information Processing Systems*, 2020.

[2] Z. Ji *et al.*, "Survey of hallucination in natural language generation," *ACM Computing Surveys*, vol. 55, no. 12, 2023.

[3] J. Maynez, S. Narayan, B. Bohnet, and R. McDonald, "On faithfulness and factuality in abstractive summarization," in *Proc. ACL*, 2020.

[4] M. Belyi, R. Friel, S. Shao, and A. Sanyal, "Luna: A lightweight evaluation model to catch language model hallucinations with high accuracy and low cost," in *Proc. COLING 2025: Industry Track*, 2025. arXiv:2406.00975.

[5] Á. Kovács and G. Recski, "LettuceDetect: A hallucination detection framework for RAG applications." arXiv:2502.17125, 2025.

[6] C. Niu, Y. Wu, J. Zhu, S. Xu, K. Shum, R. Zhong, J. Song, and T. Zhang, "RAGTruth: A hallucination corpus for developing trustworthy retrieval-augmented language models," in *Proc. ACL (Volume 1: Long Papers)*, pp. 10862–10878, 2024.

[7] P. Manakul, A. Liusie, and M. J. F. Gales, "SelfCheckGPT: Zero-resource black-box hallucination detection for generative large language models," in *Proc. EMNLP*, 2023.

[8] H. Sankararaman, M. N. Yasin, T. Sorensen, A. Di Bari, and A. Stolcke, "Provenance: A light-weight fact-checker for retrieval augmented LLM generation output," in *Proc. EMNLP 2024: Industry Track*, 2024.

[9] I. Poey, J. Liu, and Q. Zhong, "RAGulator: Lightweight out-of-context detectors for grounded text generation," in *Proc. EMNLP 2025: Industry Track*, pp. 1057–1071, 2025. arXiv:2411.03920.

[10] L. Bergeron, I. Buhnila, J. François, and R. State, "HalluGuard: Evidence-grounded small reasoning models to mitigate hallucinations in retrieval-augmented generation," in *Findings of ACL*, 2026. arXiv:2510.00880.

[11] Y. Zha, Y. Yang, R. Li, and Z. Hu, "AlignScore: Evaluating factual consistency with a unified alignment function," in *Proc. ACL (Volume 1: Long Papers)*, 2023.

[12] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence embeddings using Siamese BERT-networks," in *Proc. EMNLP-IJCNLP*, 2019.

[13] W. Wang *et al.*, "MiniLM: Deep self-attention distillation for task-agnostic compression of pre-trained transformers," in *Advances in Neural Information Processing Systems*, 2020.

[14] J. Chen *et al.*, "BGE M3-Embedding: Multi-lingual, multi-functionality, multi-granularity text embeddings through self-knowledge distillation." arXiv:2402.03216, 2024.

[15] K. Cho, B. van Merriënboer, C. Gulcehre, D. Bahdanau, F. Bougares, H. Schwenk, and Y. Bengio, "Learning phrase representations using RNN encoder–decoder for statistical machine translation," in *Proc. EMNLP*, 2014.

[16] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *Proc. ICLR*, 2019.

[17] C. Szegedy *et al.*, "Rethinking the Inception architecture for computer vision," in *Proc. IEEE CVPR*, 2016.

7 Conflict of Interest Statement

The author declares that there are no conflicts of interest regarding the publication of this paper.

A Settings of the Final Configuration

Table 8 lists the main settings of the final model.

Table 8. Main settings of the final model (values from the run configuration files; configuration E uses the same values without the eight verifier features).

Setting	Value	Setting	Value
Embedding dimension	64	Dropout / spatial / word	0.4 / 0.2 / 0.1
Word-level GRU hidden size	64	Label smoothing	0.05
Sentence-context GRU	32 per direction	Weight averaging (EMA)	0.999
Evidence chunks (top-k)	3 (at most 100 tokens)	Optimiser	AdamW
Maximum sentence length	64 tokens	Learning rate / weight decay	2×10^{-3} / 0.01
Vocabulary: minimum count	3	Learning-rate patience	2 evaluations
Batch size	about 128 sentences	Evaluation interval / patience	200 steps / 8 evaluations
Token loss weight / positive cap	0.5 / 10	Maximum epochs	30
Response loss weight / positive weight	0.5 / 1.25	Selection criterion	validation AUPRC
Sentence encoder	all-MiniLM-L6-v2	Seeds (final model)	0–9
Sentence-level features	11 grounding + 8 verifier	Ensemble threshold (10 seeds)	0.474

B Development Path and Secondary Experiments

The neural configurations below were explored in sequence during development. They differ from one another in several settings at once, so the rows of Table 9 are not single-factor ablations; only F is the final model. (A) uses basic token features and no span loss. (B) adds the six grounding groups and the token loss. (C) adds to B the regularisation of Appendix A and four further feature groups. (D) uses the six groups, the regularisation, the token loss, the response-level loss, the sentence-context layer, synthetic hard negatives and longer data-to-text evidence (top six chunks, at most 240 tokens). (E) is D without the synthetic negatives and without the longer evidence. (F), the final model, is E with the eight verifier features. F was trained with ten seeds, E with five and every other configuration with three.

Table 9. Development path, test split. "F1 over seeds" is mean ± s.d. of single networks; the remaining columns refer to seed 0. ‡Earlier runs not covered by the results file used for the main tables.

Configuration	Seeds	F1 over seeds	F1 (95% CI)	P	R	AUROC	AUPRC
(A) Basic token features‡	3	0.670 ± 0.011	0.677 (0.649–0.707)	0.622	0.743	0.831	0.757
(B) A + six groups + token loss‡	3	0.695 ± 0.014	0.700 (0.675–0.727)	0.626	0.794	0.847	0.780
(C) B + regularisation + four extra groups	3	0.708 ± 0.006	0.707 (0.680–0.734)	0.658	0.764	0.841	0.759
(D) Six groups + all training additions	3	0.702 ± 0.007	0.701 (0.674–0.727)	0.630	0.789	0.850	0.783
(E) D without synthetic data and long evidence	5	0.723 ± 0.003	0.719 (0.696–0.744)	0.646	0.810	0.863	0.793
(F) E + verifier features (final model)	10	0.726 ± 0.007	0.722 (0.697–0.746)	0.664	0.791	0.863	0.789

From C to E. Configuration C, with regularisation and four extra feature groups but no response loss, context layer or synthetic data, reached a mean F1 of 0.708 ± 0.006. Configuration D, which adds the structural changes, did not improve on it (0.702 ± 0.007). Removing two of D's additions produced E, which is clearly better (0.723 ± 0.003). E also differs from C in a second respect: the response loss and the context layer are switched on and the four extra feature groups switched off. No controlled ablation was run to show which of these changes matters, so the contributions of the response-level loss, the context layer and the four dropped feature groups remain unattributed.

Size of the sentence encoder. With all other settings of D unchanged, replacing MiniLM by the much larger bge-m3 encoder [14] gave a mean F1 of 0.699 ± 0.011 against 0.702 ± 0.007 over three seeds, a seed-0 AUROC of 0.847 against 0.850, a seed-0 AUPRC of 0.778 against 0.783 and a validation AUPRC of 0.831 against 0.832. Each run also took roughly two to four times longer in our logs, so we keep MiniLM.

Longer data-to-text evidence. Giving data-to-text sources six chunks (up to 240 tokens) instead of three changed the mean F1 from 0.706 (E with synthetic negatives) to 0.702 (D) and the seed-0 data-to-text AUROC from 0.840 to 0.835, so we dropped this setting.

Per-task thresholds. Tuning thresholds per task on validation data lowered the ensemble's F1 in all seven configuration families of the earlier ablations, by 0.018 to 0.059. With only 251 validation sources and a hallucination rate that differs from the test set's, there may be too little data for such thresholds to transfer, but we did not test this. A single shared threshold is used throughout.

Stacking. Stacking (a logistic regression over ten statistics of the sentence logits and a task indicator, fitted on validation responses with source-grouped out-of-fold scores) was run only for five-seed ensembles in earlier experiments. Compared with the plain five-seed ensemble it raised AUROC by 0.005 and AUPRC by 0.006–0.008 (E: 0.869 to 0.874 and 0.802 to 0.809; F: 0.873 to 0.878 and 0.805 to 0.813) and left F1 within ± 0.006. Across seven configuration families in the earlier ablations, the AUROC gain was always 0.005–0.007. Stacking was not repeated for the final ensemble and is fitted on a validation set whose hallucination rate differs from that of the test set, so it is excluded from the final model.